INCH-POUND
MIL-S-19678A(OS)

20 May 1988

SUPERSEDING
ML-S-19678(NOrd)
6 December 1956

MILITARY SPECIFICATION

SLIP RING ASSEMBLY MARK 7 MOD 0
(GUN FIRE CONTROL SYSTEM MARK 68)

This specification is approved for use within the Naval Sea Systems command, Department of the Navy and is available for use by all Departrnents and Agenies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for a slip ring assembly designated as Slip Ring Assembly Mark 7 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards forma part of this specification to the extent specified herein. Unless othewise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SFECIFICATIONS

FEDERAL

QQ-S-571	Solder Tin Alloy, Tin-Lead Alloy, and Lead Alloy
TT-C-490	Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-I-6181	interference Control Requirements, Aircraft Equipment
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-E-16400	Electronic, Interior Communication, and Navigation Equipment Naval Ship and Shore, General Specification
MIL-W-16878	Wire, Electrical, Insulated, General Specification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (Code 3730), Indian Head, MD 20640-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A. FSC 1210

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MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-F-18870	Fire Control Equipment, Naval Shipboard, General Specification for

STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-1303	Painting of Naval Ordnance Equipment

2.1.2 Other Government documents drawings and publications The following other Government documents, drawings, and publlcations form a part of this specification to the extent specified herein. Unless othewise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

BUREAU OF ORDNANCE

1373457	Wiring Diagram of Slip Ring Mark 2
1468511	Wiring Diagram of Slip Ring Mark 2
1814580	Requisition Drawing Showing Mounting Dimensions of the Slip Ring Assembly Mark 7 Mod 0

PUBLICATIONS

BUREAU OF ORDNANCE

CS 6341	Miscellaneous General Ordnance Design requirements
OD 3000	Lubrication of Ordnance Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B-633-78 Zinc on Iron and Steel, Electrodeposited coatings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through Libraries or other informational services).

2.3 Order of Precedence. In the event of a confiict between the text of this specification and the references cited herein, the text of this specification shall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. A sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 General. The slip rings are required to provide electrcal connection between circuits of the revolving element of the gun director and the fixed ship's structure in order to obtain unlimited train rotation of the director. The overall size and mounting dimensions shall conform to Bureau of Ordnance Drawing 1814580. The slip ring covered by this specification consists of the following major components:

a. A junction box secured at the top of the revolving element of the slip ring assembly it encloses the terminal boards and marker strips for all circuits leading to the revolving element.

b. A tube rotor assembly bolted to the bottom of the junction box and extending down through the center of the slip ring assembley

c. A center section housing subassembly enclosing the tube and rotor assernbly, slip rings, and stationary brush block assemblies.

d. A waveguide assembly extending from the top of the junction box down through the center of the tube rotor assembly to the bottom of the slip ring assembly. Both ends of the waveguide extend through a seal at the top and bottom of the assembly.

e. Stationary brush block assemblies engaging the rotating slip rings connected to vertical supports on the housing subassembly.

3.3 Materials and Component Parts.

3.3.1 Materials. The choice of materials used in the construction of the equipment shall be consistent with good engineering practice and the requirements imposed by this specification for each application. Selection should be made from the specifications listed in Section 2.1 insofar as may be practicable to do so.

3.3.2 Slip rings. The material used for manufacture of the slip rings shalli be solid coin silver, consisting of 90 percent sliver and 10 percent copper ± 1 percent.

3.3.2.1 Slip ring hardness. The slip rings shall have a hardness of Brinell 125 minimum.

3.3.3 Shock mounts. Shock mounts shall be provided to mount the stationary housing to the ship's deck. A freedom of 3 inches vertical and 3 inches horizontal shall be provided. Also, shock mounts shall be provided for mounting the slip ring assembly to the gun director drive tube The configuration of the slip ring assembly required for mounting the slip ring assembly to the gun director shall conform to Bureau of Ordnance Drawing 1814580.

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3.3.4 Flammable materials. Materials which may be ignited or exploded by an electrical spark shall not be used. Flammable materials impregnating the potting compounds may be used if they are enclosed in fire resistant containers

3.3.5 Toxic materials. Materials which may produce harmful toxic effects under conditions that occur in naval vessels, including fire shall not be used.

3.3.6 Die castings. Die castings shall not be used without specific approval of the Naval Sea Systems Command for each part or piece.

3.3.7 Magnesium. Magnesium shall not be used without specific approval of the Naval Sea Systems Command for each part or piece. This does not preclude use of alloys containing a low percentage of magnesium.

3.3.8 Aluminum alloys. Insofar as possible, aluminum alloys shall be used wherever weight saving can be effected without sacrificing other essential qualities. Aluminum alloys shall conform to MIL-F-18870.

3.3.9 Iron and steel. The use of iron and steel, except where required for electro-magnetic or low-frequency shielding purposes, shall be held to a minimum commensurate with strength requirements. Cast iron shall not be used without specific approval of the Naval Sea Systems Command for each individual application. Steel alloys of the corrosion-resistant type shall be used where strength and corrosion resistance are required in the same part.

3.3.10 Insulating materials. All insulating materials shall have high arc resistance and low moisture absorption. Ceramic, glass, teflon, nylon base plastic or other materials approved for the purpose shall be used where the impressed electrical potential exceeds 500 volts rms ac or equivalent dc.

3.3.11 Hook-up-wire. The hook-up-wire for the internal wiring shall be in accordance with the requirements of MIL-W-16678 (NAVY). Extruded insulation in accordance with this specification shall be used.

3.4 Construction.

3.4.1 Welding. In the event welded fabrications are employed in the equipment, the welds shall be of ample size with thorough penetration and good fusion, and shall be substantially free of injurious or unsightly surface defects. All welds shall be capable of withstanding strength tests to destruction in which case failure shall not occur within the weld itself. There shall be no deleterious effects of the welding process in adjacent material. All welds shall be cleaned of scale oxidation products and excess flux by sandblasting, wirebrushing, or other suitable means. Acid, if used for cleaning, shall be neutralized and removed. Tack or spot welds, if employed, shall be provided in sufficient number to ensure adequate strength; a minimum of two such welds shall be used to hold a part.

3.4.2 Soldering. Only Class H solder in accordance with QQ-S-571 shall be used for electrical connections. All connections shall be mechanically secure before soldering. Only the minimum of solder necessary for a good electrical connection shall be used, and care shall be exercised to ensure that adjacent parts or wiring are not damaged in the process. Corrosive flux shall not be used and the flux residue shall be removed when soldering is completed.

3.4.3 Electrolytic action. The faying surfaces of dissimilar metals shall be suitably treated to inhibit electrolytic action in presence of atmospheric sodium chloride in those cases where such action would normally occur.

3.4.4 Moisture pockets. Recesses in which condensed moisture can collect shall be avoided or properly drained.

3.4.5 Protective coatings and treatment.

3.45.1 General. All parts manufactured from materials not resistant to corrosion shall be provided with a protective coating of finish, except as noted hereinafter. The coatings need not be applied to:

a. Mechanisms that are bathed in oil or coated with grease such as bearings, gears, and cams.

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b. Interior surfaces of items such as tube and coil shields, or potted or hermetical sealed units.
c. Standard or commercial parts which already have a suitable protective coating.

3.4.5.2 Ferrous metal parts Surfaces of ferrous metal parts that are to be painted shall be cleaned and phosphate treated in accordance with TT-C-490. The surface produced shall be Type 1 as described therein.

3.4.5.3 Aluminum alloy parts. The surface of aluminum alloy parts shall be anodized in accordance with MIL-A-8625. Surfaces to be painted maybe given a chemical treatment in accordance with MIL-C-5541 in lieu of the anodic coating.

3.4.3.4 Painting. Painting shall be in accordance with MIL-STD-1303

3.4.5.3 Plating. When required and permitted, the plating of metal surfaces shall be in accordance with the applicable specification listed herein.

3.4.5.5.1 Plating instructions. The following restrictions on the use of plating apply.

a. Cadmium plating shall not be used in enclosed spaces where flaking wiII cause malfunctioning of the equipment.

b. Silver plating shall not be used where severe arcing is likely to occur.

c. Zinc plated parts shall be given a subsequent chromate treatment in accordance with ASTM B633, Type II

d. Bright plating, or brightly plated parts, shall not be used exterior to the principal enclosure

3.5 Environmental Requirements.

3.5.1 Temperature. The equipment shall be capable of satisfactory operation between the temperature limits of -20°F and 150°F.

3.5.2 Shock. The equipment shall be capable of withstanding the vibration requirements of MIL-STD-167. Type I tests, as specified therein, shall apply.

3.5.3 Vibration. The equipment shall be capable of withstanding the vibration requirements of MIL-STP-167. Type I tests, as specified therein, shall apply.

3.5.4 Degree of enclosure. The entire slip ring assembly shall be spraytight as defined in MIL-STD-108

3.5.5 Humidity. The equipment shall operate under humidity conditions set forth in MIL-E-16400.

3.5.6 Moisture, fungus, and corrosion resistance. The slip ring assembly shall be moisture, fungus, and corrosion resistant as required by NAVORD OS 6341.

3.6 Electrical requirements of slip rings

3.6.1 R. F. section. Three coaxial conductors as follows:

1 60 MC IF
1 MOD Trigger
1 IFF Sync

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3.6.2 High voltage section.

4 Rings -7800 volts dc, 5 Ma.

3.6.3 Control section.

240 Rings -115 volts ac, 5 amps
2 Rings -300 volts dc 5 amps
28 Rings - Group Shielded -115 volts ac, 5 amps
(Static shields without brushes)

3.6.4 Shielded section.

24 Shielded pair - Group shielded 115 volts , 5 amps
2 Brushed ground rings
9 Unshielded rings - One ring adjacent to each shielded pair to carry shields of 9 shielded pair

3.6.5 Power section.

12 Rings - (440 volts, ac or dc, 50 amp peak, 25 amp average)

NOTE:

The wiring of the Slip Ring Assembly Mark 2 shown on Bureau of Ordnance Drawings 1373457 and 1468511 shall be followed insofar as practicable to minimize changes to the director wiring.

3.6.6 Resistance. The contact resistance at rated current measured from brush to ring shall not exceed 0.005 ohm. The resistance shall not vary in excess of 0.005 ohm at rotational speeds of 10 revolutions per minute.

3.6.7 Wear resistance. Brush dust shall not cause breakdown by decreasing the creepage path during life of the brushes which shall be 2000 hours of operation.

3.6.8 Cross talk. The maximum cross talk for the various circuits shall be as follows:

a. Coaxial Circuits - 45 db at 60 mc and 70 ohm load
b. Control Section - 60 db at 400 cps and 220 ohm load
c. Shielded Section - 40 db at 150 kc at 220 ohm load

3.6.9 Radio interference. Interference shall not be permitted in the frequency range of 0.15 to 500 mc.

3.6.10 Safety. The equipment design shall incorporate methods to protect operating and servicing personnel from accidental contact with electrical potentials in excess of 70 volts. The outside enclosure of a unit shall be routed to a single ground connection; the subassembly chassis shall not be used for this purpose. The ground terminal on all subassemblies shall be routed to a single ground terminal within the outside enclosure; the enclosure itself shall not be used in lieu thereof. A suitable ground strap shall be provided to electrically ground the outside enclosure to ship structure. The corners and edges of the outside enclosure shall be rounded to prevent contact injury to personnel.

3.7. Lubrication. Lubrication shall be in accordance with Publication OD 3000.

3.8 Maintenance parts. Maintenance parts for the slip ring assembly shall be furnished as specified in the contract or purchase order.

3.9 Drawings publications and specifications. Drawings, publications, and specifications shall be furnished in accordance with MIL-F-18670.

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3.10 Workmanship. All parts comprising the complete equipment shall be manufactured and finished in a thoroughly workmanlike manner. Particular attention shall be given to maintaining a high degree of reliability, safety, and appearance. Precautions shall be observed throughout manufacture and assembly to ensure a finished product free of burrs, chips, process residues, or other foreign material that would adversely affect appearance or operation.

4. QUALITY ASSURANCE PROVSIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contractor purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contractor purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection= forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions Unless otherwise specified (see 6.2.1), inspection conditions shall be as specified in the applicable test method (see 4.5.3).

4.4 First article inspection. First article inspection shall be peformed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on a sample assembly which has been produced with equipment and procedures normally used in production.

4.4.1 Sampling. The first article sample shall be subjected to all the tests of 4.5.3 The sample shall be a first article assembly submitted by the contractor prior to the initiation of production.

4.4.2 Failure. Failure of a first article slip ring assembly to comply with any of these test requirements shall result in withholding of first article approval.

4.5 Quality conformance inspection. All quality conformance test samples shall be treated as first article samples (see. 6.3). Quality conformance inspection shall consist of the tests specified in 4.5.3.

4.5.1 Sample size. The tests specified in 4.5.3 shall be performed throughout production on 1 unit randomly selected from each group of 10 assemblies submitted for Government acceptance Reduced testing of 1 assembly randomly selected from each group of 50 or less assemblies submitted for Government inspection may be instituted following compliance with all quality conformance test requirements of 2 sample assemblies selected from 2 consecutively submitted groups of 10 assemblies. Normal testing of 1 in 10 ring assemblies shall be reinstated upon the failure of an assembly to comply with any of the quality conformance test requirements.

4.5.2 Failure. Upon failure of a quality conformance sample Government inspection of other tests may continue: however, acceptance will be withheld pending correction of the defects responsible for the rejection. The procuring activity shall be notified of accepted lots possibly containing similar defects. When the cause of rejection has been determined and corrective action within the limitation of the design disclosure taken by the contractor to the satisfaction of the Government Inspector, then production maybe continued and acceptance inspection resumed. In the event the cause of rejection cannot be determined, the procuring activity shall be so advised.

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4.5.3 Tests

4.5.3.1 Dielectric. All electrical circuits shall be capable of passing the dielectric strength test described below. All circuits in the slip ring assembly shall be subjected to high potential tests of 500 volts ac for less than 500 volts operating voltage; and twice operating plus 1000 volts for other circuits.

4.5.3.2 Insulation resistance. Insulation resistance tests shall be made to determine the insulation resistivity. The insulation resistance shall be 100 megohms minimum at 60 percent relative humidity.

4.5.3.3 Radio interference. The assembly shall be tested for radio interference in accordance with MIL-I-6181.

4.5.3.4 Rotation test. The slip rings shall be rotated at rated speed (10 rpm) for 24 hours, during which-time visual inspection will be made to determine the amount of sparking, brush dust, or other inherent defects in the operation of the slip ring assembly.

4.6 Inspection of packaging. The sampling and inspection of the packaging and marking shall be in accordance with the requirements of MIL-E-17555.

5. PACKAGING

5.1 Packaging. The equipment shall be packaged and prepared for domestic shipment in accordance with MIL-E-17555.

5.2 Marking. In addition to any special marking required by the contractor purchase order all shipping containers shall be marked in accordance with MIL-STD-129 see (6.2.1).

6. NOTES

6.1 Intended use. The slip rings are intended to provide electrical connection between circuits of the revolving element of the gun director and the fixed ship's structure in order to obtain unlimited train rotation of the director.

6.2 Ordering data.

6.2.1 Acquisition requirement. Acquisition documents should specify the following:

a. Title number, revision, and date of this specification

b. When reduced quality conformance testing is allowed (see 4.5.1)

c. If the contractor is not responsible for inspection (see 4.1)

d. If the contractor should not use his own laboratory or other suitable facility for performance of inspection requirements (see 4.1)

e. Inspection conditions, if other than as specified (see 4.3)

f. Any special packaging and marking requirements (see 5.1 and 5.2).

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6.4 Patent notice. When Government specifications, publications, drawings, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said specifications, publications, drawings, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing.

Brush block
Dielectric strength
Fire control
Gun director
Junction box
Resistance
Slip ring
Tube rotor
Waveguide assembly

Preparing Activity
Navy OS
(1210-NO13)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19678A(OS)	SHIP RING ASSEMBLY MARK 7 MOD 0 (GUN FIRE CONTROL SYSTEM M[redacted]B)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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